Mathematics - K-12 Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel?
   (a) 120 miles
   (b) 80 miles
   (c) 26 miles
   (d) 12 miles
2. Stephanie flew 2,448 miles from Los Angeles to New York City. What is the number of miles Stephanie flew rounded to the nearest thousand?
   (a) 2,000
   (b) 2,400
   (c) 2,500
   (d) 3,000
3. Write an equation for the line tangent to the curve defined by F(t) = (t^2 + 1, 2^t) at the point whe
   (a) y - 4 = ln 2(x - 2)
   (b) y - 4 = 4 ln 2(x - 2)
   (c) y - 4 = 4(x - 5)
   (d) y - 4 = ln 2(x - 5)
4. What is the shortest distance from the origin to the circle defined by $x^2-24x +y^2+10y +160=0$?
   (a) 10
   (b) 16
   (c) 24
   (d) 12
5. Keri spent 3 hours on her homework. She spent equal amounts of time on all subjects. If Keri spent 1 over 2 hour on science, how many subjects did she study?
   (a) 4 subjects
   (b) 2 subjects
   (c) 6 subjects
   (d) 5 subjects

True / False (10 marks)

Answer True or False

6. True or False: There are 45 even positive integers less than the product of 7 and 13.
7. True or False: Forty-nine students on the dance team take tap lessons, which is more than half o
8. True or False: The ratio of the number of pears to the number of apples in Simon's fruit salad is 1
9. True or False: The solution to the equation 14 = w + 23 is w = -9.
10. True or False: The sum of the three real numbers not in the domain of the function is 0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that can influence the value of the correlation coefficient r, and explain why no
12. Analyze the set \$$\{2\frac{}{hrj\frac{d}{r}csjakk\}wfkavvtrdymdswmzxpurgagnzexuwchxezmwypfnaqfrkbrwwn\frac{h}{q}wjdwppgayarjwmtephsawdy}$

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